

Tathagata Saha

RESEARCHER · ASTRONOMY

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Research Positions

Inter-University Centre for Astronomy and Astrophysics (IUCAA)

POST DOCTORAL FELLOW

with eROSITA German Consortium, Compact objects working group

Pune, India

15.09.2025 onwards

Leibniz Institute for Astrophysics Potsdam (AIP)

RESEARCH ASSISTANTSHIP AND GUEST RESEARCHER

with Dr. Mirko Krumpe

Potsdam, Germany

01.12.2023 - 31.05.2026

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences

PHD ASTRONOMY

Thesis Supervisor: Prof. dr. hab. Alex G. Markowitz

Warsaw, Poland

01.10.2019 - 22.03.2024

Education

Presidency University

M.Sc. PHYSICS

- Thesis: Connection Between the Accretion Disc and Corona in the Active Galactic Nuclei
Advisors: Prof. Ritaban Chatterjee, Prof. Indranil Chattopadhyay

Kolkata, India

01.08.2017 - 01.07.2019

Presidency University

B.Sc.(HONOURS) PHYSICS

- Thesis: Accretion Disc Corona Interaction in Active Galactic Nuclei
Honours thesis/undergraduate research advisor: Prof. Ritaban Chatterjee

Kolkata, India

21.07.2014-01.07.2017

Research Interests

- Multi-wavelength spectral and timing studies of supermassive black hole transients, particularly changing-look AGNs (CLAGNs) and tidal disruption events (TDEs), to probe rapid accretion state changes.
- Evolution and geometry of the circumnuclear obscuring medium (torus), its connection to accretion in-flows/outflows in AGNs and supermassive black hole growth.
- Time-resolved spectral studies of radio-quiet Seyferts to constrain the structure and evolution of accretion sub-components such as the disk, corona, and broad-line region.
- Theoretical modeling of accretion physics: disk-corona interactions, jet formation, and instabilities governing state transitions in AGNs.
- Investigating the role of accretion-driven feedback on black hole growth, galaxy evolution, and its imprint on the cosmic X-ray background and large-scale structure.

Research Skills

- X-ray and optical data reduction and analysis using classical and Bayesian methods Multinest/Ultraneest (Nested Sampling) and Markov Chain Monte Carlo in multiple frameworks including XSPEC, Sherpa, and any python environment.
- Contribution to observation proposals for optical (SALT, ESO-VLT) and X-ray telescopes (XMM-Newton, Chandra, NuSTAR, NICER, ASTROSAT) via performance of feasibility simulations and developing of scientific cases.

- Developer skills: Python (proficient) implementation of numerical methods, data wrangling (Numpy, Pandas), statistical analysis (Scipy, Scikit-Learn), and deep learning (TensorFlow).
- Academic service: Referee for the AAS, The Astrophysical Journal (ApJ)

Publications

REFEREED AND ACCEPTED PUBLICATIONS

- Saha, T** et al., eRASSU J043115.8-711730: The first pulsating symbiotic super-soft X-ray source in the Magellanic Bridge, 2026, Accepted for publication in MNRAS; 2026
- Saha, T** et al., Multiwavelength study of extreme variability in LEDA 1154204: A changing-look event in a type 1.9 Seyfert, A&A, 702, A28; 2025
- Saha, T** et al., The Close AGN Reference Survey (CARS). Long-term spectral variability study of the changing look AGN Mrk 1018; A&A, 699, A205; 2025
- Saha, T**; Markowitz, A. G.; Buchner, J, Inferring the morphology of AGN torus using X-ray spectra: a reliability study; MNRAS, 509, 5485; 2022
- Homan, D., Krumpe M., Markowitz, A., **Saha, T.**, and 16 co-authors. Discovery of the luminous X-ray ignition eRASSt J234402.9-352640; I. Tidal disruption event or a rapid increase in accretion in an active galactic nucleus?, A&A 672, A167; 2023
- Laha, S., Markowitz, A., Krumpe, M., Nikutta, R., Rothschild, R., **Saha, T.** The Variable and Non-variable X-Ray Absorbers in Compton-thin Type II Active Galactic Nuclei, ApJ, 897, 66; 2020
- Markowitz A., Krumpe M., ... **Saha, T.** and 13 co-authors. eROSITA Detection of a Cloud Obscuration Event in the Seyfert AGN EC 04570-5206, 2024, A&A, 684, A101; 2024
- Krishnan, S., Markowitz, A., ... **Saha, T.** and 13 co-authors. An X-ray flaring event and a variable soft X-ray excess in the Seyfert LCRS B040659.9-385922 as detected with eROSITA, A&A, 691, 19; 2024
- Markowitz A., Krumpe M., ... **Saha, T.** and 12 co-authors. A changing-look Seyfert discovered by eROSITA reveals a two-component broad-line region, A&A, 710, A222; 2026
- Mereghetti, S et al.....**Saha, T.** Time domain astronomy with the THESEUS satellite, Exp Astron, 52, 309; 2021

ONGOING PROJECTS

- Characterizing the stellar population in the Magellanic Bridge using eROSITA detections and multi-wavelength follow-up.
- Detection Changing Obscuration AGNs in eROSITA archives using X-ray hardness ratios.
- Development of a neural network based spectral classifier that can facilitate efficient identification of changing look AGNs by flagging spectral type changes in optical spectral surveys (e.g. SDSS).
Github repository: github.com/tathagatas1996/AGNopspecNN-base
- Development of a code base (Saha & Naddaf in prep.) to study the structural evolving of the broad line region in transient changing-look AGNs using Failed Radiatively Driven Outflow model (Naddaf et al., 2022, A&A, 663, A77)

Teaching Experience

- August 2024 **X-ray Data Analysis with XSPEC, Content:** X-ray data and statistic, Likelihood functions, Bayesian Statistics: Markov Chain Monte Carlo (MCMC), Nested Sampling, PyXSPEC ,
Location: School of Astrophysics, Presidency University, Kolkata

Scholarships & Grants

- 2023 **Research Assistant Scholarship**, Deutsches Zentrum für Luft und Raumfahrt (DLR), P.I. Dr. Mirko Krumpe, FKZ 50 OR 2004
- 2020 **Ph.D. Scholarship on Active Galactic Nuclei**, Polish National Science Center (NCN), P.I. Prof. dr. hab. Alex G. Markowitz, 2018/31/G/ST9/03224
- 2017 **Pre-Ph.D. Selection with Scholarship**, Inter-University Center for Astronomy and Astrophysics, Pune, India
- 2017 **Integrated Ph.D. Selection with Scholarship**, Department of Physics, Indian Institute of Science, Bengaluru, India
- 2017 **All India Rank 26, Joint Entrance Screening Test for Ph.D. programs**,

Presentations

Invited Oral Presentations

- January 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*. Invited talk: X-ray Spectral Fitting 2022 (XSF22), Prague, Czech Republic
- April 2023, *Active Galactic Nuclei: Active Galactic Nuclei: The multi-wavelength signature of black hole accretion*. Invited Outreach talk: Kolkata, India.
- September 2024, *Morphology of circumnuclear accreting gas in Active Galactic Nuclei*, Invited Talk, School of Astrophysics, Presidency University, Kolkata
- April 2025, *Multiwavelength study of the accretion substructures in AGN transients*, Invited Talk, CESSI, IISER, Kolkata
- July 2025, *Multiwavelength study of the accretion substructures in AGN transients*, Invited Talk, High Energy Astrophysics Group, IUCAA, Pune

Contributed Oral Presentations

- September 2021, *Determining the AGN torus morphology using X-ray Spectra: A reliability study*, Polish Astronomical Society Meeting 2021, Szczecin, Poland
- July 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*, Sexten Center for Astrophysics, Sexten, Italy
- December 2022, *A BXA-driven study of reliability of X-ray Spectral fits in determining AGN torus morphology*, TORUS2022 Conference, Leiden, Netherlands
- March 2023, *Multiwavelength follow-up studies of eROSITA detected AGN transients*, Astronomical Society of India Meeting 2023, Indian Institute of Technology, Indore, India
- June 2023, *Multiwavelength follow-up study of a flaring changing-look AGN with double-peaked Balmer lines*, SALT Conference 2023, Warsaw
- October 2023, *Multiwavelength follow-up study of a flaring changing-look AGN with double-peaked Balmer lines* Gaia Alerts Conference 2023, Malta
- July 2024, *Multiwavelength follow-up studies of eROSITA detected AGN transients*, Conference on Blazars and the Restless AGN, Presidency University, Kolkata, India
- June 2026, *J0431-71: The first symbiotic supersoft source in the Magellanic Bridge*, X-ray Universe, Elche, Spain
- July 2026, *Changing look AGNs: divergence in the observed timescales of state transition*, Conference on Blazars and the Restless AGN, IISER, Mohali, India

Contributed Poster Presentation

- July 2021, *Determination of AGN Torus Morphology with X-ray Spectra: A reliability study*, European Astronomical Society Meeting 2021
- March 2022, *Determination of AGN Torus Morphology with X-ray Spectra: A reliability study*, Astronomical Society of India Meeting 2023, Aryabhata Research Institute for Observational Sciences, Nainital, India
- July 2022, *Determination of AGN Torus Morphology with X-ray Spectra: A reliability study*, European Astronomical Society Meeting 2022
- May 2022, *Multi-wavelength study of extragalactic transients detected with eROSITA: A flaring event in an AGN*, XMM-Newton Workshop 2022, Villafranca del Castillo, Madrid, Spain

June 2023, *Multi-wavelength study of a flaring CLAGN with double-peaked Balmer lines detected with eROSITA and ZTF*, The Restless AGN, Napoli, Italy

July 2023, *Multi-wavelength study of a flaring CLAGN with double-peaked Balmer lines detected with eROSITA and ZTF*, European Astronomical Society Meeting 2023, Krakow

Mentoring

2026	Mahek Patel , Florida Institute of Technology, USA	<i>IUCAA</i>
until 2026	Ayon Mondal , University of Maryland, Baltimore County, USA	<i>AIP, IUCAA</i>
until 2026	Undergraduate Name , Presidency University	<i>AIP, IUCAA</i>

Outreach, Service, & Professional Development

January 2027	Intl. Conference: X-ray Polarization of Compact Objects , Scientific Organizing Committee	<i>IUCAA, Pune, India</i>
June 2023	Intl. Conference: SALT 2023 Conference, Warsaw , Local Organizing Committee	<i>CAMK-PAN, Warsaw, Poland</i>
2021- 2023-	Theseus Mission , Contributing Scientist, changing look AGNs with the Theseus Mission	
	Astronova Club, IEEE, Calcutta University , Membership	<i>Kolkata, India</i>